THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

PMBA6089 Big Data Consumer Analytics

GENERAL INFORMATION			
Instructor	Dr. Tuan Q. Phan		
Office	KK Leung 704	**Telephone No.**	
Email	tphan@hku.hk	**Consultation times**:	After class
Pre-requisites Course Code & Title		**Co-requisites Course Code & Title**	N/A
Tutor	Jasmine Guo, jasmineguo@connect.hku.hk		
Other important details:			

COURSE DESCRIPTION

Consumers leave a massive digital trail through mobile devices, social media, transactions, and information communication technologies (ICT). Companies and organizations in all industries face this plethora of consumer data and are looking to turn it into value through advertising and better products and services. This course looks to introduce students to the various methods and strategies to make sense of the data. We will cover topics including advance and Big Data analytics, machine and deep learning, text mining and analysis, image and video analysis, social network analysis and geospatial and mobility analytics. Students will have real hands-on practice exercises with these methods, learn how to generate insights from large and unstructured data, and build an understanding of its strategic relevance in creating sustainable competitive advantage. *No programming nor coding experiences are required*. The course will also bridge these technical methods with social sciences and consumer behavior. We will cover a number of case studies, with an emphasis on Asia, in sundry industries including retailing and e-commerce, advertising, FinTech and retail financial sectors, telecommunication, and transportation and smart city. Finally, we will discuss the interwoven dependencies on the legal frameworks with Big Data, including issues relating to privacy, regulations, rights, consumer reactions and adherence to data legislation in different markets. Prior backgrounds in marketing research methods is recommended, but not required.

Students will have exclusive use to the software (Orange) on the ACRC Data Analytics Platform with real datasets and case studies to learn to interpret and build an intuition for Big Data.

COURSE OBJECTIVES

1. To gain exposure to Big Data methods for Big Data predictive analytics, text mining, image & video analysis, and spatial and mobility analytics.
2. To build an understanding the of the business value of Big Data analytics
3. To understand how Big Data can be used to better understand customer behavior
4. To gain an appreciation of how customer analytics can apply to various industries

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition and internationalization of knowledge of the programme discipline
PLO2: Application and integration of knowledge
PLO3: Inculcating professionalism and leadership
PLO4: Developing Global Outlook
PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1: Understand how to use different Big Data methods and tools for real-world problems to understand consumer behaviour	PLO2,5
CLO2: Create sustainable competitive strategies using Big Data	PLO1,2,3,4,5

CLO3: Understand the data landscape from the consumer, legal, and regulatory framework	PLO 2,3,4
CLO4: Learn about business cases which leverage Big Data-centric strategies	PLO2,3,4,5
CLO5: Explain how Big Data helps to build insights into social sciences, consumer behaviour, and economics.	PLO1,2,3,5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	**Expected contact hour**	**Aligned Course Learning Outcomes**
T&L1. Interactive Lectures & Case discussions	30	CLO1,2,3,4,5
T&L2. Case reading & preparation	25	CLO2,4,5
T&L3. Textbook and course reading material	25	CLO1,2,4,5
T&L4. Exercises	40	CLO1,5
Total	**120**	

COURSE ASSESSMENT METHODS

Assessment Methods	**Brief Description (Optional)**	**Weight**	**Aligned Course Learning Outcomes**
A1. Class participation	Participating in class activities including discussions of assignments, taking pre-class surveys and activities, attendance, and name cards.	30	CLO1,2,3,4,5
A2.Case writeup and discussion	In-class discussion of cases. Prepare for cold-calls. Short writeup of cases.	30	CLO2,3,4,5
A3.Exercises	Submit written exercises and assignments in groups.	40	CLO1,5
Total		**100%**	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	Consistently demonstrate a thorough grasp of the subject as evidenced by achieving an outstanding performance in understanding of marketing principles and concepts, critical analysis and synthesis, application of knowledge, formulation of marketing solutions, written and verbal communication and teamwork skills.
B+, B, B-	Frequently demonstrate a substantial grasp of the subject as evidenced by achieving a proficient performance in understanding of marketing principles and concepts, critical analysis and evaluation, application of knowledge, formulation of marketing solutions, written and verbal communication and teamwork skills.
C+, C, C-	Occasionally demonstrate a general grasp of the subject as evidenced by achieving a moderate performance in understanding of marketing principles and concepts, analysis and evaluation, application of knowledge, formulation of marketing solutions, written and

	verbal communication and teamwork skills.
D+, D	Demonstrate a partial grasp of the subject as evidenced by achieving an adequate performance in understanding of marketing principles and concepts, analysis and evaluation, application of knowledge, formulation of marketing solutions, written and verbal communication and teamwork skills.
F	Demonstrate a poor grasp of the subject as evidenced by achieving a poor performance in understanding of marketing principles and concepts, analysis and evaluation, application of knowledge, formulation of marketing solutions, written and verbal communication and teamwork skills.

Assessment Rubrics for Each Assessment

A1. In-class participation (30%) & A2. Case discussion (30%)

A+, A, A-	B+, B, B-	C+, C, C-	D+, D	F
Extremely well prepared for class discussion, active in sharing views and attended at least 90% of classes.	Partially prepared for class discussion, quite active in sharing views and attended at least 80% of classes.	Not well prepared for class discussion, limited active in sharing views and attended at least 80% of classes.	Not well prepared for class discussion, no sharing of views and attended at least 70% of classes.	Never prepared for class discussion and no sharing of views, experience, and attend less than 70% of classes.

A3. Exercises (40%)

A+, A, A-	B+, B, B-	C+, C, C-	D+, D	F
• Demonstrate a strong understanding of all relevant knowledge •Present arguments that have an element of originality •Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills	•Demonstrate a good understanding of all relevant knowledge •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills	•Demonstrate a basic understanding of the concepts involved •Present arguments in a well-structure manner •Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills	•Demonstrate a minimum understanding of the concepts involved •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills	•Demonstrate a poor understanding of the concepts involved •Present arguments poorly •Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Lecture	Date	Lecture Topic	Reading Material/Discussions	Assignment/Due
1	Saturday AM Jan. 10	• Course Introduction • Introduction to (Big Data) Analytics • Introduction to the software (Orange) on the ACRC Data Analytics Platform • Analytics Review • Regressions Introduction Machine Learning o Clustering		
2	Saturday PM Jan. 10	• *Introduction to Text Mining* • Sentiment Analysis		• **Survey 1 due before class** • Assignment #1,

		• Social Media analysis		Text Analysis, available
3	Saturday AM Jan. 17	• Case 1 discussion • Assignment 1 in class	• Case 1	• **Assignment #1 in class** • Assignment #2 available
4	Saturday PM Jan. 17	• *Introduction to Images & Video Analysis* • Classification and predictions • Prediction, machine learning, deep learning • Case 2 discussion	• Case 2	
5	Saturday AM Jan. 24	• Advertising, eye tracking, customer tracking, and applications • Panel data • Panel models • Causal analysis		• **Survey 2 in class** • **Assignment #2 in class**
6	Saturday PM Jan. 24	• *Introduction to Social Network Analysis* • Influence, communities, and the value of Social Network Data • Case 3 discussion	• Case 3	• Assignment #3, Social Networks, available
7	Saturday AM Jan. 31	• *Introduction to location analysis* • Location technologies GIS, indoor & outdoor location • Assignment #3 in class		• **Assignment #3 in class**
8	Saturday PM Jan. 31	• Trajectories • Geofencing • Applications of location • Case #4 discussion	• Case 4	• Assignment #4, Location analysis, available
9	Saturday AM Feb. 7	• *Privacy* • Assignment #4 in class		• **Assignment #4 in class**
10	Saturday PM Feb. 7	• Assignment #4 discussion • Wrap up	• Case 5	

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

The SFTL questionnaire

COURSE POLICY

The University Regulations on academic dishonesty will be strictly enforced! Please check the University Statement on

plagiarism on the web: http://www.hku.hk/plagiarism/.	
ADDITIONAL COURSE INFORMATION	
This course will utilize MOODLE.	